

SIMATS ENGINEERING
ASSESSMENT TOOL 1

J. Manoj

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COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO5:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL5)

Assessment Tool Weightage: 40%

QUESTIONS: 3

Annexure A

82%

Total Marks:30

Constraint-Based Problem Solving

1. A multinational company is opening a new branch office in a remote rural location where high-speed wired internet is unavailable. The office must support cloud-based applications, video conferencing, and secure communication with the headquarters. The company has a limited budget for connectivity but requires uninterrupted business operations. As a network engineer, recommend a suitable internet connection, networking devices, and security mechanisms. Justify your recommendations by considering bandwidth limitations, reliability, cost, and future scalability
2. A university plans to provide campus-wide wireless internet access for more than 3,000 students across classrooms, laboratories, libraries, and hostels. The network must support a large number of simultaneous users while minimizing signal interference and ensuring secure access. Due to budget constraints, only a limited number of wireless access points can be deployed. Analyze the requirements and propose an efficient wireless network design, including access point placement, frequency band selection, and authentication methods. Explain how your solution balances coverage, performance, security, and cost.
3. A financial institution is upgrading its internal computer network to improve cybersecurity while allowing employees to access sensitive customer information and online banking services. The organization must comply with strict security policies without significantly affecting network performance or increasing operational costs. Design a secure network architecture by selecting appropriate network segmentation techniques, firewall deployment, authentication methods, and intrusion detection mechanisms. Justify your design by considering the constraints of security, performance, compliance, and budget.

82

24-5

SIMATS ENGINEERING

ASSESSMENT TOOL 1

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COURSE OUTCOME COVERED

co5: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)
Assessment Tool Weightage: 30%

QUESTIONS: 5

Annexure A

Total Marks:50

Code of Conduct:

I __j.manoj(192572143)__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

j.manoj

QUESTION 1

A multinational company is opening a new branch office in a remote rural location where high-speed wired internet is unavailable. The office must support cloud-based applications, video conferencing, and secure communication with the headquarters. The company has a limited budget for connectivity but requires uninterrupted business operations. As a network engineer, recommend a suitable internet connection, networking devices, and security mechanisms. Justify your recommendations by considering bandwidth limitations, reliability, cost, and future scalability.

1. Problem Analysis

The company wants to open a branch office in a remote rural area. In this area, wired broadband (fibre/leased line) is not available. Still, the office must run cloud applications, video calls, and secure communication with the head office (HQ), and the company does not want the network to go down. The budget is limited, so the solution must be cost-effective but reliable, and it should be possible to upgrade the network later as the branch grows.

2. Constraints Identified

Constraint	Explanation	Impact on Design
No wired/fibre internet	Remote rural location has no ISP fibre or leased-line coverage	Must use wireless WAN technology (satellite / 4G-5G)
Cloud apps + video conferencing	Needs consistent bandwidth and low latency	Requires QoS and a backup link
Secure link to HQ	Sensitive business data travels over public internet	Site-to-site VPN with strong encryption is mandatory
Limited budget	Company cannot spend heavily on links or hardware	Prefer 4G/5G LTE over costly VSAT satellite; use SMB-grade devices
Uninterrupted operations	Business cannot tolerate downtime	Dual-WAN failover (primary + backup ISP)
Future scalability	Branch may grow (more staff, more apps)	Choose devices/links that support capacity upgrade without full replacement

3. Problem Decomposition (Sub-Problems)

1. Choosing the right last-mile internet connection for a rural site.
2. Selecting networking devices to connect the office LAN to the internet.
3. Designing a reliable and redundant WAN link (avoiding single point of failure).
4. Securing data in transit between branch and HQ.
5. Keeping the whole solution within a limited budget while allowing future growth.

4. Proposed Internet / Transmission Media

Since fibre/wired broadband is not available, the best cost-reliability balance is a Fixed 4G/5G LTE wireless connection as the PRIMARY link, backed by a low-cost VSAT satellite link as the SECONDARY (backup) link.

Option	Bandwidth	Reliability	Cost	Verdict
Wired Leased Line / Fibre	High (dedicated)	Very High	Very High + long rural rollout time	Not available / not feasible here
4G/5G Fixed Wireless (LTE router)	25–150 Mbps typical	High (depends on tower coverage)	Low monthly cost, cheap CPE	RECOMMENDED as Primary link
VSAT Satellite	5–20 Mbps, higher latency (~600 ms)	High, works anywhere	High equipment + monthly cost	RECOMMENDED as Backup link only
Dial-up/Radio link	Very Low	Low	Low	Rejected — cannot support video/cloud apps

Reasoning: A 4G/5G connection gives enough bandwidth for cloud apps and HD video conferencing at a low monthly cost, which directly satisfies the budget constraint. Because rural mobile towers can occasionally have outages, a VSAT satellite link is kept only as a backup — it is expensive per Mbps, so using it as primary would break the budget constraint, but as an automatic failover it protects the 'uninterrupted operations' constraint at low extra cost.

5. Networking Devices Required

Device	Purpose	Why Needed
4G/5G LTE Wireless Router (Primary WAN)	Converts mobile network signal into office LAN internet	Only viable broadband source in rural site
VSAT Satellite Modem (Backup WAN)	Secondary internet path	Provides redundancy for reliability constraint
SD-WAN capable / Dual-WAN Router	Automatically switches between 4G and VSAT	Ensures zero/near-zero downtime (auto-failover)
UTM Firewall (Unified Threat Management)	Combines firewall + VPN + antivirus + IDS in one box	Cost-effective security for a small branch (one device instead of many)
Managed Switch (8–24 port, Gigabit, PoE)	Connects PCs, IP phones, and Wi-Fi AP inside office	Local LAN connectivity with VLAN support for scalability
Wireless Access Point (Wi-Fi 6)	Wireless connectivity for staff devices	Flexible, low-cost internal coverage; easy to add more later
UPS (Uninterruptible Power Supply)	Backup power for router/switch/firewall	Rural areas often face power cuts — protects uptime

6. Security Mechanisms

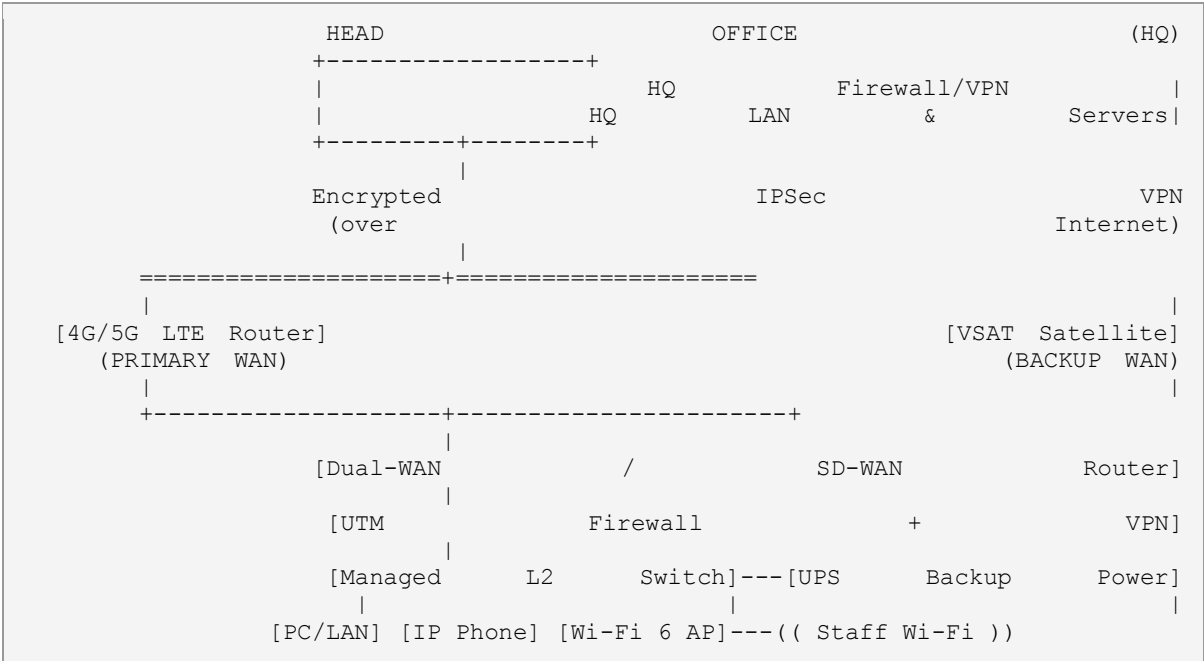
- Site-to-Site IPSec VPN between branch UTM firewall and HQ firewall — encrypts all traffic between office and HQ so cloud/video/data stays private over the public 4G/VSAT link.
- UTM Firewall with stateful packet inspection — blocks unauthorized inbound traffic from the internet.
- WPA3-Enterprise on Wi-Fi with a RADIUS/username-password login — prevents unauthorized devices from joining office Wi-Fi.

- Endpoint Antivirus + regular patching on all branch PCs — reduces malware risk since malware can enter through the internet link.
- Role-based access control (RBAC) for cloud application logins — limits what each employee can access.
- Scheduled encrypted backups of local data to HQ/cloud — protects business continuity if the branch link fails.

7. Proposed Network Architecture (Step-by-Step)

6. Internet enters the branch through the 4G/5G LTE router (primary) and the VSAT modem (backup), both connected to a dual-WAN/SD-WAN router.
7. The dual-WAN router constantly checks both links; if the 4G link fails, traffic automatically shifts to VSAT within seconds.
8. The dual-WAN router connects to the UTM Firewall, which inspects all traffic and terminates the IPSec VPN tunnel to HQ.
9. The firewall connects to a managed Layer-2 switch, which distributes LAN connections to office PCs, IP phones, and printers.
10. A Wi-Fi 6 access point (connected to the switch, powered via PoE) gives staff wireless access with WPA3-Enterprise security.
11. Video conferencing and cloud application traffic is marked high-priority using QoS rules on the router so calls stay smooth even when other traffic (downloads, updates) is running.
12. All office-to-HQ traffic travels only through the encrypted VPN tunnel, keeping business data secure end-to-end.

8. ASCII Network Topology Diagram



9. Advantages & Disadvantages

Aspect	Advantages	Disadvantages
4G/5G Primary Link	Fast to deploy, low cost, good bandwidth for cloud & video	Depends on mobile tower signal strength/congestion

Aspect	Advantages	Disadvantages
VSAT Backup	Works almost anywhere, true redundancy	Higher latency, costlier per Mbps
UTM Firewall	Single device = lower cost & easier management	Slightly less powerful than separate dedicated appliances
Site-to-Site VPN	Strong end-to-end encryption, HQ data stays private	Adds small latency overhead

10. Cost-Conscious Justification

- 4G/5G router + data plan is far cheaper than laying a fibre line in a remote area, and there is no installation waiting time — this directly satisfies 'limited budget'.
- VSAT is used ONLY as backup (not primary), so the company avoids VSAT's high per-Mbps running cost while still getting reliability.
- A UTM firewall replaces 3–4 separate security appliances with one box, cutting both hardware cost and maintenance cost.
- Wi-Fi 6 AP and managed switch are chosen at SMB (small-medium business) grade — enough performance for a small branch, without paying for large-enterprise capacity that isn't needed yet.
- Scalability: because the switch supports VLANs and the router supports higher LTE data plans, the company can add more users/APs later just by upgrading the subscription plan — no need to redesign the network from scratch.

11. Conclusion

For a rural branch with no wired internet, a 4G/5G LTE connection as the primary link with a VSAT satellite backup, combined with a dual-WAN router, UTM firewall, and site-to-site VPN, gives the company a reliable, secure, and low-cost network. This design supports cloud applications and video conferencing today, keeps operations running even during link failure, and can scale up easily as the branch grows — meeting every constraint given in the problem: bandwidth, reliability, cost, and scalability.

QUESTION 2:

A university plans to provide campus-wide wireless internet access for more than 3,000 students across classrooms, laboratories, libraries, and hostels. The network must support a large number of simultaneous users while minimizing signal interference and ensuring secure access. Due to budget constraints, only a limited number of wireless access points can be deployed. Analyze the requirements and propose an efficient wireless network design, including access point placement, frequency band selection, and authentication methods. Explain how your solution balances coverage, performance, security, and cost.

1. Problem Analysis

The university needs Wi-Fi across a large campus (classrooms, labs, library, hostels) for more than 3,000 students who will connect at the same time. Because of budget limits, only a limited number of Access Points (APs) can be bought, so each AP must be placed and configured carefully to cover the maximum area with the least interference, while still keeping the network secure and fast.

2. Constraints Identified

Constraint	Explanation	Impact on Design
3,000+ simultaneous users	Very high device density	Need enterprise-grade APs with high user-capacity, not home routers
Multiple zones (classrooms, labs, library, hostels)	Different building types & user density per zone	AP placement and count must vary by zone, not one-size-fits-all
Limited number of APs (budget)	Cannot cover every corner with dedicated APs	Must optimize placement using building layout + prioritize high-density zones
Minimize interference	Overlapping Wi-Fi channels slow the network	Careful channel planning and dual-band (2.4/5 GHz) use
Secure access	Only authorized students/staff should connect	WPA2/WPA3-Enterprise with 802.1X + RADIUS authentication

3. Problem Decomposition (Sub-Problems)

13. Estimating how many APs are needed per zone based on user density and area.
14. Deciding AP placement to maximize coverage and minimize dead zones and overlap.
15. Choosing frequency bands (2.4 GHz vs 5 GHz vs Wi-Fi 6E) to reduce interference and boost speed.
16. Selecting an authentication method that is both secure and easy for 3,000+ students to use.
17. Balancing all of the above against a limited AP budget.

4. AP Placement Strategy by Zone

Zone	User Density	AP Placement Strategy	Frequency Priority
Classrooms	High (40–60 students/room, scheduled)	1 AP per 2 mid-size classrooms (ceiling-mounted, centered)	5 GHz primary (high speed, short dwell interference)
Laboratories	Medium-High, many wired PCs too	1 AP per lab room, mounted away from metal equipment	5 GHz (less interference from equipment)
Library	High, long user sessions, needs stability	APs every 800–1000 sq.ft, ceiling-mounted, higher AP density	Dual-band: 5 GHz primary, 2.4 GHz fallback
Hostels	High at night, thick walls between rooms	1 AP per 2–3 rooms or per floor corridor, using mesh/backhaul APs	2.4 GHz for wall penetration + 5 GHz where possible
Open areas / Corridors	Low-medium, roaming users	Fewer APs, positioned to cover walkways between buildings	2.4 GHz (longer range, lower cost)

Placement principle used: high-density, small, enclosed rooms (classrooms, labs, library) get more APs per square foot because many devices connect at once; low-density open areas get fewer, longer-range APs. This lets the limited AP budget be spent where it matters most — satisfying both the coverage and cost constraints together.

5. Frequency Band Selection

Band	Characteristics	Where Used Here
2.4 GHz	Longer range, better wall penetration, more interference, lower speed, only 3 non-overlapping channels	Hostel corridors, outdoor/open areas, fallback for older devices
5 GHz	Shorter range, less interference, much higher speed, many non-overlapping channels	Classrooms, labs, library — main band for most users
Wi-Fi 6 (802.11ax)	Uses OFDMA to serve many devices per AP efficiently, ideal for high user density	Recommended AP standard campus-wide to support 3,000+ concurrent connections with fewer APs

Using Wi-Fi 6 APs is the key decision that lets a LIMITED number of APs still serve 3,000+ students: Wi-Fi 6's OFDMA technology allows one AP to efficiently serve many simultaneous devices instead of serving them one at a time, directly solving the 'limited AP budget vs. high user count' conflict.

6. Channel Planning to Minimize Interference

- Use the standard non-overlapping channels: 1, 6, 11 on 2.4 GHz and multiple 20/40 MHz non-overlapping channels on 5 GHz.
- Assign neighboring APs (in adjacent rooms) to different, non-overlapping channels in a repeating pattern so signals don't clash.
- Reduce AP transmit power in small rooms (classrooms, hostel floors) so signals don't bleed too far and interfere with the next AP's cell.
- Enable automatic RF (radio frequency) channel and power management on the wireless controller so channels re-adjust automatically if interference is detected.

7. Authentication & Security Methods

Mechanism	Purpose
WPA3-Enterprise / WPA2-Enterprise (802.1X)	Every student authenticates with their own university ID + password (not one shared Wi-Fi password) — much more secure and lets access be revoked per-user
Central RADIUS Server	Verifies student/staff credentials against the university database before granting Wi-Fi access
Captive Portal (for guests)	Visitors get a separate, time-limited, isolated guest network
VLAN Segmentation	Students, staff, and guests are placed on separate VLANs so guest/hostel traffic can't reach admin/exam servers
Firewall + Content filtering at gateway	Blocks malicious sites and controls bandwidth abuse (e.g., limits torrenting)

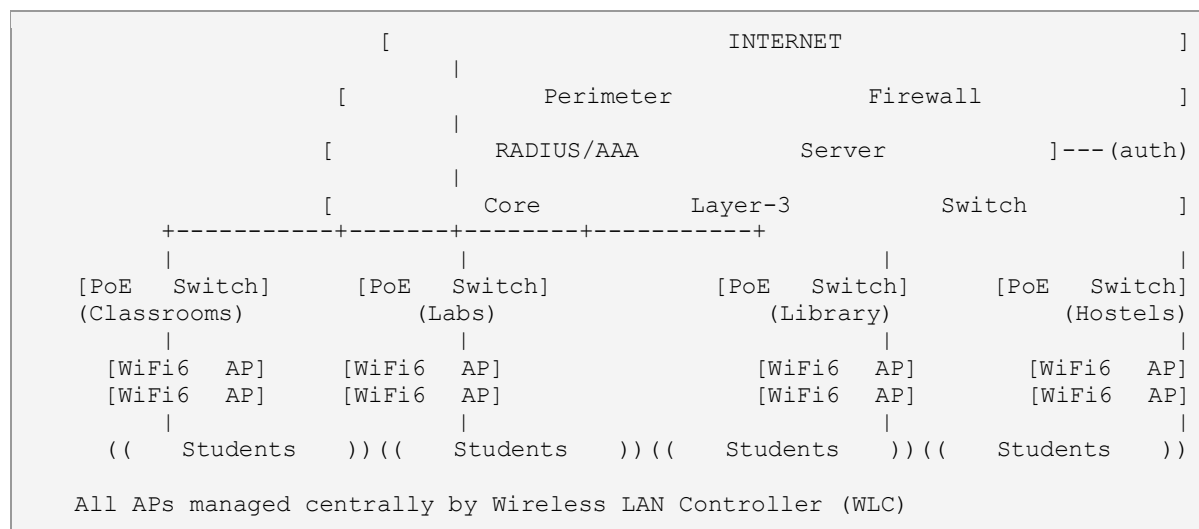
8. Networking Devices Required

- Enterprise Wi-Fi 6 Access Points (ceiling-mounted, PoE-powered) — placed per the zone table above.
- Wireless LAN Controller (WLC) — centrally manages all APs, handles channel/power auto-tuning and roaming between APs.
- PoE Switches (per building) — power and connect the APs using a single cable (saves electrical wiring cost).
- Core Layer-3 Switch — connects all building switches and routes traffic between VLANs.
- RADIUS/AAA Server — handles 802.1X authentication for all students and staff.
- Perimeter Firewall — protects the campus network from external threats and manages internet gateway traffic.

9. Step-by-Step Design Explanation

18. Survey each building (classrooms, labs, library, hostels) to note size, wall material, and expected user count.
19. Calculate AP count per zone using the rule: high-density small rooms get 1 AP per room/2 rooms; low-density open areas share fewer APs.
20. Choose Wi-Fi 6 APs so each AP can efficiently handle more simultaneous devices, keeping total AP count within budget.
21. Assign 5 GHz as the primary band in classrooms/labs/library for speed; use 2.4 GHz for hostel wall-penetration and open areas for range.
22. Plan non-overlapping channels in a repeating pattern and enable auto RF management via the Wireless LAN Controller.
23. Connect all APs to PoE switches, which connect to a core switch, which connects to the RADIUS server and firewall/gateway.
24. Configure WPA2/WPA3-Enterprise with 802.1X so each student logs in with their own university credentials.
25. Segment traffic using VLANs — Student VLAN, Staff VLAN, Hostel VLAN, Guest VLAN — for security and to control bandwidth per group.

10. ASCII Network Topology Diagram



11. Advantages & Disadvantages

Aspect	Advantages	Disadvantages
Wi-Fi 6 APs (fewer, smarter APs)	Fewer APs needed = lower cost, handles high user density well	Higher per-unit cost than older Wi-Fi 5 APs
Zone-based placement	Budget spent where user density is highest	Requires careful site survey beforehand
802.1X + RADIUS	Strong per-user security, easy to revoke access	Slightly more complex initial setup than a shared password
VLAN segmentation	Isolates guest/hostel traffic from critical systems	Needs Layer-3 switch configuration knowledge

12. Cost-Conscious Justification

- Wi-Fi 6's OFDMA lets each AP serve more students at once, so the university needs fewer total APs to cover 3,000+ users — this is the main way the design meets the budget constraint.
- PoE switches remove the need for separate electrical wiring to each AP, cutting installation cost.
- Zone-based placement avoids wasting APs in low-traffic open areas, saving budget for high-density rooms.
- A central Wireless LAN Controller reduces IT staffing/management cost by automating channel and power tuning instead of manual per-AP tuning.

13. Conclusion

By deploying Wi-Fi 6 access points using a zone-based placement strategy (more APs in dense classrooms/library, fewer in open areas), applying careful 2.4/5 GHz channel planning, and securing access with WPA2/WPA3-Enterprise 802.1X authentication through a RADIUS server, the university can serve 3,000+ simultaneous students with minimal interference and strong security — all while keeping the total number of access points, and therefore the cost, within budget.

QUESTION 3:

A financial institution is upgrading its internal computer network to improve cybersecurity while allowing employees to access sensitive customer information and online banking services. The organization must comply with strict security policies without significantly affecting network performance or increasing operational costs. Design a secure network architecture by selecting appropriate network segmentation techniques, firewall deployment, authentication methods, and intrusion detection mechanisms. Justify your design by considering the constraints of security, performance, compliance, and budget.

1. Problem Analysis

A bank/financial institution needs to upgrade its internal network so it is much more secure, because employees handle sensitive customer data and online banking services. At the same time, the new design must follow strict compliance rules (like PCI-DSS for card data), must not slow the network down, and must not raise operating costs much. So the design has to balance security, performance, compliance, and budget together.

2. Constraints Identified

Constraint	Explanation	Impact on Design
Strict security required	Sensitive customer & banking data at risk	Multi-layer defense: segmentation + firewall + IDS/IPS + strong auth
Regulatory compliance	Financial data is regulated (e.g., PCI-DSS, banking cyber norms)	Logging, encryption, access control, and audit trails are mandatory
No major performance drop	Employees need fast access to banking systems	Use hardware-based/next-gen firewalls and place IDS inline efficiently, not everywhere
No big cost increase	Operational budget is limited	Reuse existing infrastructure where possible; choose scalable, licensed-as-needed devices

3. Problem Decomposition (Sub-Problems)

26. Segmenting the network so sensitive systems (customer data, banking servers) are isolated from general employee/office traffic.

- 27.Deciding where and how to deploy firewalls to control traffic between segments and the internet.
- 28.Choosing authentication methods strong enough for compliance but not disruptive to daily work.
- 29.Placing intrusion detection/prevention systems to catch attacks without slowing the network.
- 30.Keeping the entire design inside a limited operating budget.

4. Network Segmentation Design

Segment (VLAN/Zone)	Contains	Access Rules
DMZ (Demilitarized Zone)	Online banking web/app servers facing customers	Only allows specific inbound ports from internet; isolated from internal LAN
Core Banking / Database VLAN	Customer data, transaction databases	Accessible ONLY from application servers and authorized admin workstations
Employee/Office VLAN	Staff PCs, email, general applications	Internet + limited internal access, no direct DB access
Management VLAN	Network device admin interfaces (routers/switches/firewalls)	Accessible only to IT/network admins via secure jump host
Guest/Branch VLAN (if applicable)	Visitor or low-trust devices	Internet-only, fully isolated from all internal segments

Segmentation is done using VLANs on Layer-3 switches with strict inter-VLAN routing rules (a form of the Zero Trust idea: nothing talks to anything else unless explicitly allowed). This means even if one segment (e.g., employee PCs) is compromised, the attacker still cannot directly reach the core banking database — this is the single biggest step for both security and compliance.

5. Firewall Deployment Strategy

Firewall Location	Type	Purpose
Perimeter (Internet edge)	Next-Generation Firewall (NGFW)	Filters all inbound/outbound internet traffic; deep packet inspection, application control
Between DMZ and Internal LAN	Internal segmentation firewall	Ensures a breach on the public-facing banking server cannot reach internal systems
In front of Core Banking VLAN	Dedicated internal firewall / ACLs on core switch	Extra protection layer for the most sensitive data (defense-in-depth)

Using a layered firewall approach (perimeter + internal + core) instead of just one big firewall means that even if outer defenses are breached, inner layers still protect the database — this directly satisfies 'strict security' without needing to buy one extremely expensive high-end firewall, which helps the budget constraint too.

6. Authentication Methods

- Multi-Factor Authentication (MFA) for all employees accessing banking applications or the VPN — password + OTP/authenticator app.
- Role-Based Access Control (RBAC) — a teller, a loan officer, and an IT admin each get different, minimum-necessary access levels (principle of least privilege).

- Active Directory / LDAP with centralized password policy — enforces strong passwords and account lockout, required for most compliance frameworks.
- Certificate-based authentication for site-to-site VPNs between branches and the data center.
- Session timeout & automatic logoff on banking applications after inactivity — reduces risk from unattended terminals.

7. Intrusion Detection / Prevention Mechanisms

- Network IDS/IPS (Intrusion Detection/Prevention System) placed inline at the perimeter and at the DMZ boundary — inspects traffic patterns and blocks known attack signatures in real time.
- Host-based IDS on core banking servers — monitors for unusual file/process activity directly on the most sensitive machines.
- SIEM (Security Information and Event Monitoring) tool collecting logs from firewalls, IDS, and servers — required for compliance audit trails and helps detect slow, distributed attacks.
- Regular vulnerability scanning and patch management schedule — required by most financial compliance standards.

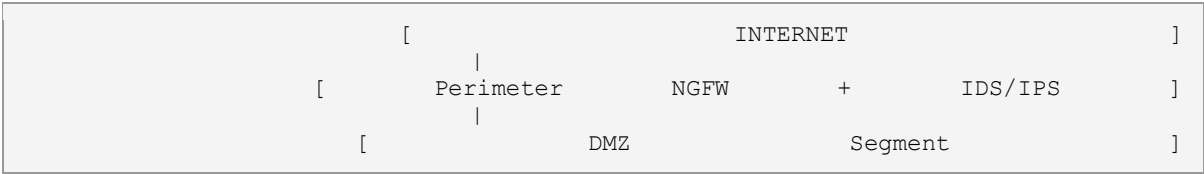
8. Networking Devices Required

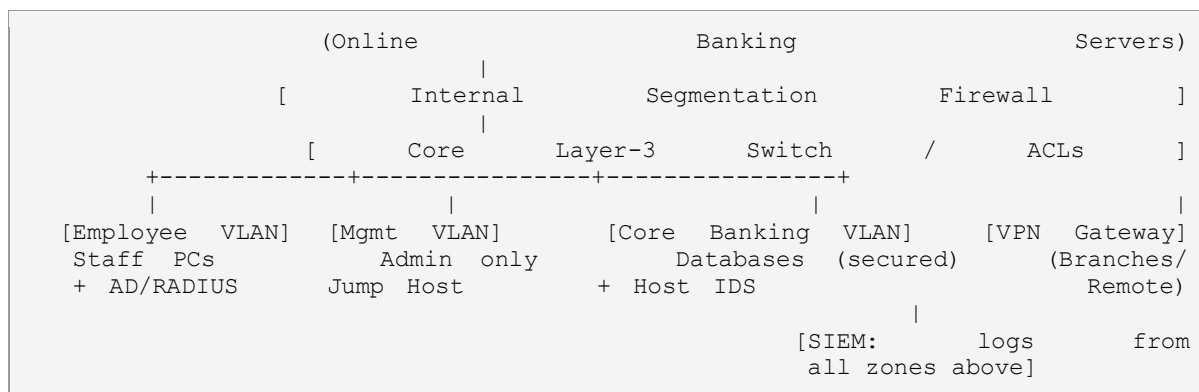
- Next-Generation Firewall (perimeter) — internet-facing protection with deep packet inspection.
- Internal segmentation firewall / Layer-3 core switch with ACLs — enforces VLAN-to-VLAN rules.
- IDS/IPS appliance (inline at perimeter and DMZ) — real-time attack detection and blocking.
- RADIUS/AAA + Active Directory server — centralized authentication for staff.
- SIEM server — centralized log collection and compliance reporting.
- VPN concentrator/gateway — secure encrypted access for remote/branch staff.

9. Step-by-Step Design Explanation

- 31. Classify all systems into segments: DMZ (public banking servers), Core Banking VLAN (databases), Employee VLAN, Management VLAN.
- 32. Deploy an NGFW at the internet edge to filter all external traffic before it reaches the DMZ.
- 33. Place a second (internal) firewall between the DMZ and internal LAN so that even a compromised public server cannot directly reach internal systems.
- 34. Configure the core Layer-3 switch with strict inter-VLAN ACLs so only authorized servers/admins can reach the Core Banking VLAN.
- 35. Deploy inline IDS/IPS at the perimeter and DMZ boundary to catch attacks in real time.
- 36. Set up Active Directory + RADIUS with MFA and RBAC so every employee logs in with their own strong, monitored, least-privilege account.
- 37. Feed logs from firewalls, IDS/IPS, and servers into a SIEM system for continuous monitoring and compliance reporting.
- 38. Schedule regular patching and vulnerability scans to keep the whole system compliant and current.

10. ASCII Network Topology Diagram





11. Advantages & Disadvantages

Aspect	Advantages	Disadvantages
VLAN Segmentation	Contains breaches, satisfies compliance separation rules	More complex initial configuration and documentation
Layered Firewalls (NGFW + internal)	Defense-in-depth; breach in one layer doesn't expose everything	Slightly more devices to manage/patch
MFA + RBAC	Strong protection against stolen passwords, meets compliance	Small extra login step for employees
IDS/IPS + SIEM	Real-time detection, full audit trail for regulators	Requires monitoring staff/time to review alerts

12. Cost-Conscious Justification

- VLAN segmentation is done mostly through configuration on existing Layer-3 switches, not new hardware — a low-cost way to massively boost security and compliance.
- A layered firewall design uses one strong NGFW at the edge plus lighter internal firewalls/ACLs, instead of multiple expensive NGFWs everywhere — controls cost while keeping defense-in-depth.
- Centralizing authentication (AD + RADIUS + MFA) is a one-time setup that then scales to all employees without extra per-user hardware cost.
- Inline IDS/IPS is placed only at key boundaries (perimeter + DMZ) rather than on every internal link, keeping both cost and performance impact low while still protecting the most important paths.
- SIEM logging (often available as a scalable/cloud-tier service) directly satisfies compliance audit requirements without needing a large in-house security team from day one.

13. Conclusion

By segmenting the network into DMZ, Core Banking, Employee, and Management VLANs, deploying a layered firewall approach (perimeter NGFW + internal segmentation firewall), enforcing MFA with RBAC for authentication, and placing IDS/IPS with SIEM logging at key boundaries, the financial institution achieves strict security and regulatory compliance. Because segmentation reuses existing switches, firewalls are layered rather

than duplicated everywhere, and IDS/IPS is placed only at critical points, the design also avoids major performance loss or cost increase — satisfying all four constraints: security, performance, compliance, and budget.